

Assessing the water metabolism of coastal urban areas based on the water mass balance framework across time periods: A case study of Cape Town, South Africa

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ABSTRACT

The status of water resources in many coastal cities has become increasingly fragile. In a changing climate, the amount and intensity of rainfall have continued to decline, causing some coastal regions to experience severe drought and deteriorated water supply situation. This article presents a comprehensive study of the urban water metabolism of Cape Town, South Africa. A water mass balance framework was utilized to examine the city's water system across four distinct periods. In addition, the water-energy nexus before and after the drought crisis was evaluated. The key findings include a large amount of surface runoff in the Cape Town area that was not utilized before the drought crisis, resulting in a natural loss potential of water supply that is 2.54 times greater than water used for supply systems. During the drought, per capita water consumption dropped by a substantial 25%; at the same time, the total rate of water loss experienced a substantial decrease of approximately 21%. Since the pandemic, Cape Town's water resources policy has shifted to diversifying water sources, and the use of wastewater and natural water losses will be optimized by more than 32% in 2040 to achieve a water-sensitive city. Future research should explore the temporal and spatial dynamics of urban water flows, the impact of socioeconomic factors, and the integration of water system optimization models for enhanced urban water management.

1. Introduction

Because of the combined effects of population growth, urbanization, and climate change, resource shortages in coastal areas have caused increasing environmental, ecological, social, and economic problems (Adrianto et al., 2021; Paul et al., 2018). Due to special geographical locations and development of coastline tourism and shipping, rapid economic development has also brought hidden problems to some coastal cities, including a single industry, a large gap between the rich and poor, and excessive resource demands (Cazcarro et al., 2022; Liu et al., 2019a; Nematian, 2023; Tian et al., 2023). In particular, water is an essential raw material that supports both residents' lives and industrial operations, bridging the natural environment with human activities (Baydaroğlu et al., 2023; Chen et al., 2020; Geleta et al., 2023). It cannot be ignored that the status of water resources in many coastal cities has become increasingly fragile. In a changing climate, the amount and

intensity of rainfall have continued to decline, causing some coastal regions to experience severe drought (Burls et al., 2019; Chen et al., 2021; Laino and Iglesias, 2023). Simultaneously, the water supply situation has further deteriorated because the water storage capacity of local dams cannot meet the rapidly growing needs of citizens and industries in coastal areas (Nhamo and Agyepong, 2019).

As coastal cities face water resource challenges, in order to have a clearer understanding of the current status of urban water resources, water managers and planners need to first integrate all water flows into an integrated system (Goswami et al., 2020; Liu et al., 2019b). At the same time, exploring the natural water cycle and the possibilities for efficient water use in coastal areas requires quantifying the performance of the entire water system (Rogers et al., 2020). In this case, the quantification of urban water performance is an evolving science and has been applied to coastal areas (Behzadian and Kapelan, 2015; He et al., 2021; Venkatesh et al., 2014). However, compared with inland cities,

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the water systems of coastal cities are often more complex due to the influence of the ocean (Griggs and Reguero, 2021). Existing water system assessment models for coastal areas often focus only on flows entering water management facilities and ignoring water sources and natural losses (Zipper et al., 2020). Moreover, water systems can interact with other urban systems, such as energy systems and agricultural systems, and this impact is not usually considered in previous assessments.

To fill existing gaps in the research, this study established a water–energy nexus in special circumstances by comparing water-related electricity consumption and energy emissions before and after a drought period to analyze and determine the correlation between the two types of flows. A water balance model and a water metabolism assessment framework were applied to demonstrate the performance of Cape Town's water system over four different periods. The findings of this study can help reveal the role of water-sensitive interventions during critical periods in achieving water performance targets, such as increasing precipitation use efficiency, diversifying alternative water sources, and improving the health of urban water metabolism, and managing increasingly stressed urban water systems in coastal cities under climate change.

2. Material and methods

2.1. Study area

As the legislative capital of South Africa, the City of Cape Town is constantly changing, including its population, economy, urban areas, and rural periphery (Savelli et al., 2023). Due to the changing climate, the capacity of the dam, which supplies water to Cape Town, fell by approximately 21% in 2018 compared with the 2014 level (Warner and Meissner, 2021). In early 2020, while Cape Town was recovering from the drought crisis, the COVID-19 pandemic began to spread across the globe, reaching Cape Town in March of that year (Hussey et al., 2021; Wang et al., 2021). Emergency measures during the pandemic altered residential mobility, consumer behavior, and material flow management, as well as patterns of wastewater, waste, and exhaust emission (Cai et al., 2021; Tian et al., 2021; Wan et al., 2024; Wang et al., 2023). In this context, water demands resulting from home confinement and increased medical needs, as well as higher volumes of medical wastewater, have led to further complex challenges for Cape Town's water supply system.

2.2. Urban water metabolism evaluation framework

The water balance framework based on urban metabolism theory and water-sensitive city vision has been applied to coastal cities in Australia and South Korea (Renouf et al., 2018; Jeong and Park, 2020). Water-sensitive cities are a manifestation of sustainable water management, typically referring to cities that act as water catchments, providers of ecosystem services, and owners of multi-source water supply solutions (Rentachintala et al., 2023; Renouf et al., 2017; Rogers et al., 2020; Serrao-Neumann et al., 2019). Additionally, they focus on promoting the contribution of water infrastructure development and equipment application to urban livability, resilience, and sustainability (Cabañas-Melo et al., 2022; Wong et al., 2013).

The urban water–energy nexus is considered important for managing urban resource scarcity crises, especially in dynamic coastal cities such as Cape Town (Wang et al., 2017; Parker, 2021). Combining this nexus with the metabolism framework can provide important guidance for sustainable urban planning and development (Fan et al., 2019; Satour et al., 2023). The methodology used in this study consisted of five parts. First, urban boundaries and water bodies were defined to determine their specific scope. Second, all required water, energy, and emissions flows were collected and estimated. All water flows were then applied to the water mass balance formula adapted from Farooqui et al. (2016) and to the actual situation in Cape Town. In addition, the rainwater

contribution formula was balanced. When both formulas were balanced, the elements in the formulas were used to evaluate the water system, including rainwater utilization for water supply, urban water metabolism performance, and hydrological performance. Moreover, data on electricity and energy emissions were used to evaluate the water–energy nexus. The framework of this study is shown in Fig. 1.

2.3. Urban system boundary

The system boundaries of this study were set as the administrative boundaries of Cape Town, including both urban and rural areas. The five main water storage dams and their locations are shown in Fig. 2. Data collected from the affiliated climate monitoring stations were included in the water flow considered in this study. Two major aquifers were within the boundaries of the waterbody study: Cape Flats and Atlantis. In addition, one minor aquifer, the Table Mountain Group aquifer, was included (Atkins et al., 2021). In this study, these aquifers served as sources of centralized groundwater abstraction and as the destination of groundwater infiltration. Furthermore, the seawater used in the desalination process is taken from the sea area belonging to the city.

This study focused on four separate timeframes aligned with the effects of climate change, drought, and the COVID-19 pandemic to examine water metabolism in Cape Town during distinct periods. Specifically, the period from 2009 to 2012 was the baseline; the period from 2015 to 2018 included the drought crisis, which was marked by a significant reduction in water inflow, such as precipitation, thereby exacerbating water scarcity during that time; the period from 2020 to 2023 comprised the COVID-19 pandemic, during which economic and social activities significantly declined; and the period from 2037 to 2040 was projected based on the water-sensitive goals and visions of Cape Town, considering the effects of several factors, such as population growth, urbanization, climate change, and the use of desalination technology (City of Cape Town, 2023; Mahmud et al., 2023). Specific data are shown in Table 1.

2.4. Flow estimation

In this study, data on the natural water flows in this work, including precipitation (P), surface runoff, and surface evapotranspiration, are collected from climate monitoring stations. In this case, surface runoff refers to stormwater runoff discharged from areas above the ground surface, which is usually measured at nodes along the river channel. Because the data detected by monitoring stations varied, surface runoff and evapotranspiration were combined into a single flow, termed natural losses (ER) in this study. Cape Town, located in a semi-arid climate, typically experiences significant natural water losses (Naik and Abiodun, 2019). Because of the complex composition of settlements in Cape Town and the lack of relevant data, groundwater infiltration (G) was obtained using the precipitation balance formula shown in Table S1.

Bulk water supply (Csw) refers to the large-scale movement of water from its source, such as reservoirs or dams, to a treatment facility or directly to distribution systems. In Cape Town, the centralized water supply system is managed by the authorities. Centralized groundwater abstraction (Cg) involves the systematic extraction of groundwater from aquifers through wells or boreholes managed by the Cape Town authority. Centralized desalination (Cd) is the process of removing salt and other minerals from seawater to produce fresh water. At present, there are ongoing investigations regarding site selection, feasibility, and water quality for a desalination plant with a daily capacity of 50 million to 150 million liters, and production is expected to begin in 2026 (City of Cape Town, 2019).

In Cape Town, there are 24 formal wastewater (W) treatment facilities that accept wastewater from different urban sectors, which is discharged into the ocean after treatment (City of Cape Town, 2021). In this case, recycled water (Rw) is separated from wastewater streams and treated for industrial, commercial, and landscaping purposes.

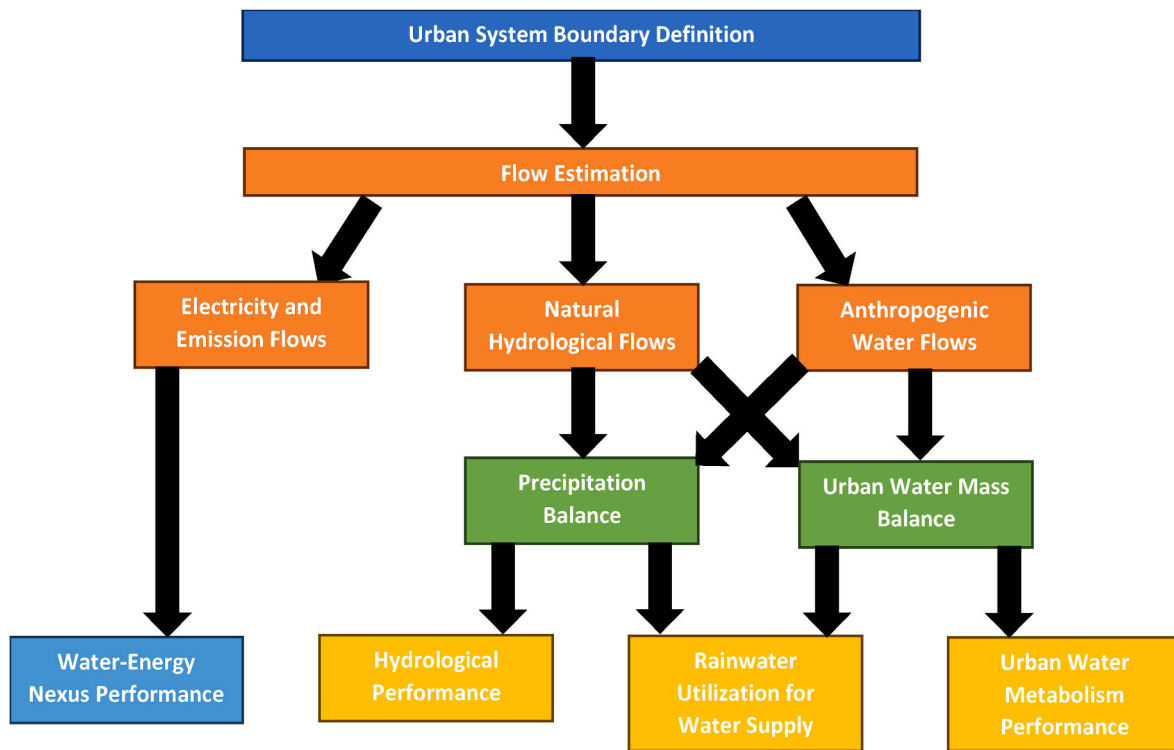


Fig. 1. Urban water metabolism evaluation framework with water-energy nexus.

Decentralized water supplies (D) are local water systems that are independent from centralized water supply networks, which are often used in rural areas or individual facilities. They include rainwater harvesting, local wells, and small-scale water treatment systems. Known losses (Cufw) refer to water that is lost from the supply network due to known reasons, such as pipe leaks, maintenance activities, or meter inaccuracies. Although Cape Town's authorities have made great progress on the issue of pipe leaks, water losses in various processes cannot be completely avoided (City of Cape Town, 2023).

2.5. Urban water mass balance

Building on the framework established by Kenway et al. (2011) and considering the actual conditions in Cape Town, the overall water balance used in this study is represented by Equation (1).

$$Csw + Cd + Cg + D = W + Cufw + Rw + S \quad (1)$$

Csw: Bulk water supply
 Cd: Centralized desalination
 Cg: Centralized groundwater abstraction
 D: Decentralized water supply
 W: Wastewater
 Cufw: Known losses
 Rw: Recycled water
 S: Water stored in urban system

Equation (1) not only provided a comprehensive view of the water metabolism in Cape Town, but also achieved the water quality balance shown in Fig. 3, ensuring that the total inflow equaled the total outflow plus any changes in the water stock.

The total centralized water supply (C) is defined in Equation (2), which will be further utilized in the water metabolism evaluation framework.

$$C = Csw + Cd + Cg \quad (2)$$

C: Total centralized water supply
 Csw: Centralized bulk water supply
 Cd: Centralized desalination
 Cg: Centralized groundwater abstraction

In this study, precipitation (P) was hypothesized to comprise components of bulk water supply (Csw), evapotranspiration, runoff (ER), decentralized water supplies (D), and groundwater infiltration (G). These components were included in Equation (3) based on their direct and intrinsic links to precipitation. The bulk water supply (Csw) was primarily derived from precipitation accumulated in surface water bodies. Evapotranspiration (ER) comprises the water lost to the atmosphere from surfaces and vegetation, a process directly influenced by the climatic conditions affected by precipitation. Decentralized water supplies (D) captured precipitation at the community level, while groundwater infiltration (G) accounted for precipitation that permeated soil to recharge the aquifers.

$$P = Csw + ER + D + G \quad (3)$$

Csw: Bulk water supply
 ER: Evapotranspiration and surface runoff
 D: Decentralized water supply
 G: Groundwater infiltration

Because Cape Town has a large number of informal settlements, and many areas lack water meters that are formally installed by Cape Town's authorities, point-to-point flow monitoring was unfeasible. Moreover, these areas usually do not have climate monitoring stations. This gap in the study data caused imbalances in the water balance equation. Therefore, water stock (S) was introduced, which became a crucial element in this study because it aided the adjustment of these disparities and enhanced the understanding of Cape Town's water metabolism within the proposed evaluation framework. When urban water metabolism was measured completely and accurately, the value of S was zero. Equation (4) was used to determine the stock of urban water.

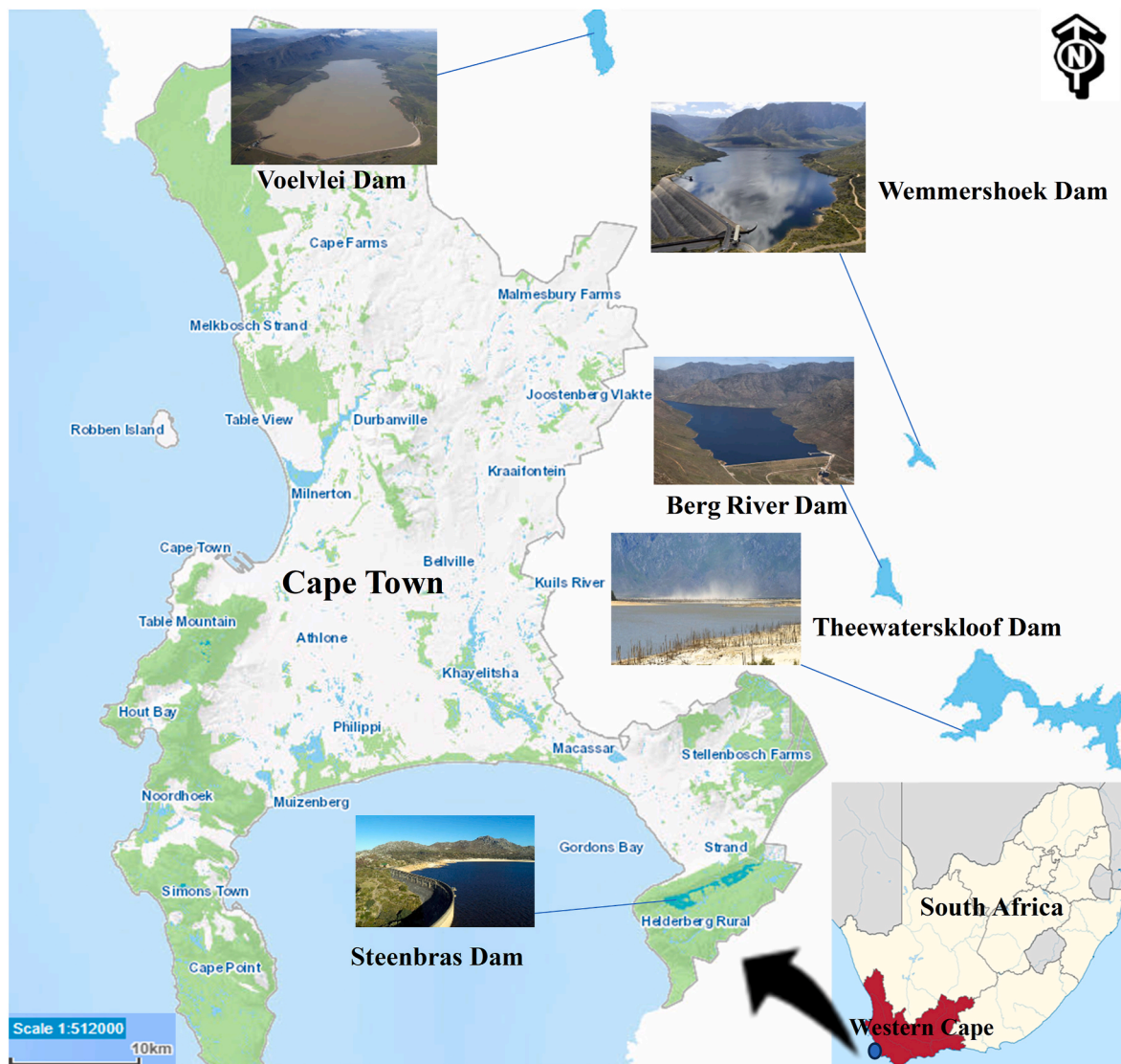


Fig. 2. Cape Town map with five main dams for water supply.

Table 1
Characteristics of the four study periods.

Characteristics	2009–2012	2015–2018	2020–2023	2037–2040
Land area ^a (km ²)	2,461	2,461	2,461	2,461
Population ^b (capita)	3,652,250	4,264,000	4,754,750	6,000,000
Major water sources	Dam water	Surface water	Dam water	Complex
Special situations	Normal	Drought crisis	COVID-19	Water sensitive

^a From Urban metabolism: A review with reference to Cape Town (Currie et al., 2017).

^b From South Africa metro area population website (Macrotrends, 2024).

$$S = Csw + Cd + Cg + D - W - Cufw - Rw \quad (4)$$

2.6. Urban water metabolism performance

Table 2 shows the urban water metabolism performance indicators and formulas that were applied to define the overall water system

metabolism of Cape Town. As described in Section 2.4, in the city's water metabolism system, rainwater recharges urban water usage by entering the centralized surface water supply system and the community-based decentralized water supply system. Relevant performance indicators were used to measure the proportion of rainwater that had been fully utilized to determine the potential utilization of rainwater. Wastewater supply potential was used to evaluate the capacity and feasibility of using treated wastewater, also known as recycled or reclaimed water, to supplement centralized water sources. This assessment is crucial in areas like Cape Town, which faces water scarcity and seeks to enhance water sustainability. The natural loss potential for water support aimed to assess the amount and reliability of natural water loss that can be effectively captured, stored, and added to the water supply system in the Cape Town area. In this study, natural water losses included evapotranspiration and surface runoff. In other words, when the potential for wastewater and natural loss was higher, it indicated more policy, behavioral, and technological gaps in realizing the effective use of wastewater and rainwater.

In this study, other urban water metabolism performances included urban water utilization, water supply internalization, and comparisons with hydrological performance during drought crises. The total number of centralized and decentralized water supplies available per capita was used to measure urban water utilization. Water supply internalization

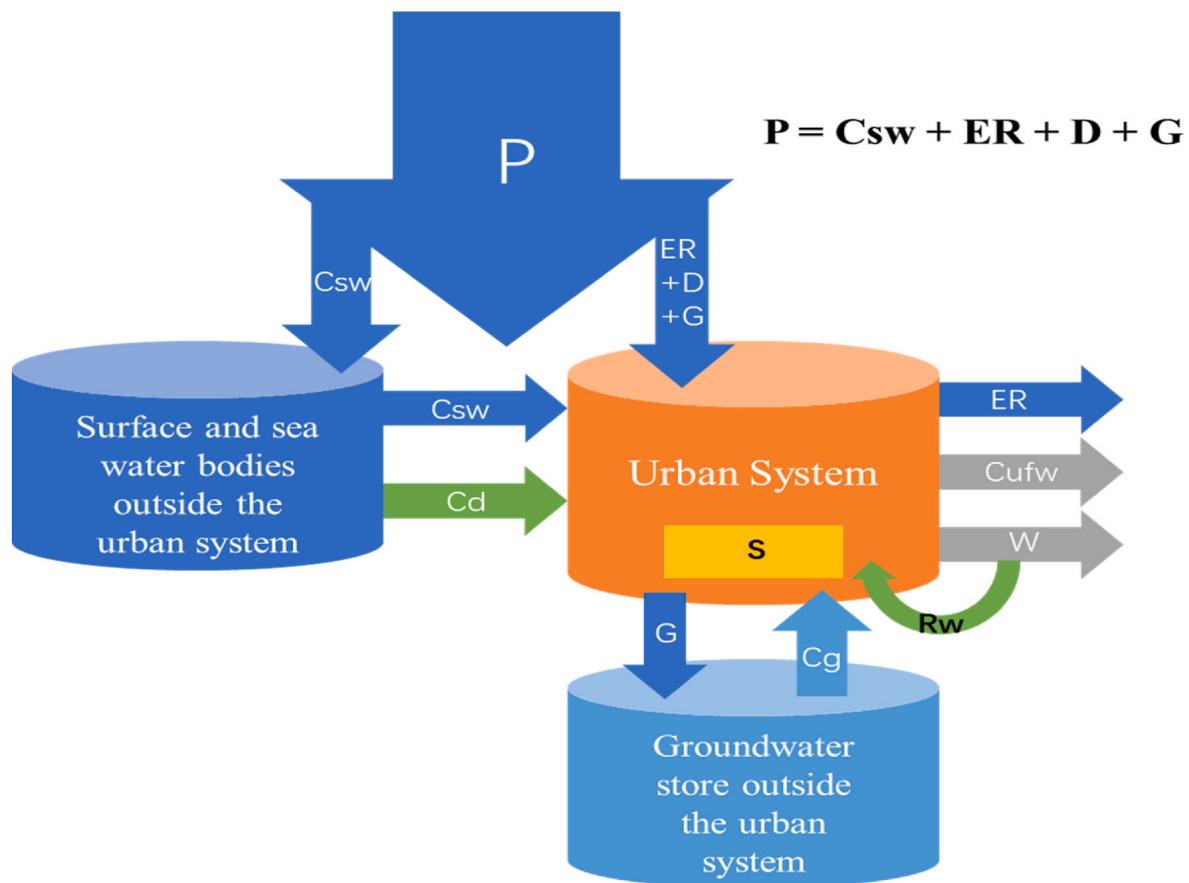


Fig. 3. Water mass balance framework.

Table 2
Urban water metabolism indicators.

Indicators	Description	Formulas
Rainwater Utilization for Water Supply		
Centralized supply use rate (%)	Bulk water supply/total precipitation	$C_{sw}/P \times 100\%$
Decentralized supply use rate (%)	Decentralized water/total precipitation	$D/P \times 100\%$
Total supply use rate (%)	Total supply/total precipitation	$(C_{sw} + D)/P \times 100\%$
Urban Water Metabolism Performance		
Urban water utilization (kL/capita/annual)	Total water use per capita per year	$(C + D + R_w)/Pop$
Water supply internalization (%)	Water demand met by decentralized supply and recycled water	$(D + R_w)/(C + D + R_w) \times 100\%$
Wastewater potential for water supply	Ratio of wastewater to water supply	$W/(C + D + R_w)$
Natural Loss Potential for water supply	Ratio of natural loss to water supply	$ER/(C + D + R_w)$
Hydrological Performance		
Hydrological performance compared with drought period-Natural loss	Ratio of post- (o) to pre-drought (d) annual flows of natural loss (ER)	ER_o/ER_d
Hydrological performance compared with drought period-Groundwater recharge	Ratio of post- (o) to pre-drought (d) annual flows of groundwater (G)	G_o/G_d

was a measure of Cape Town's development of alternative water sources based on the ratio of decentralized water supply and recycled water to total water consumption. Hydrological performance evaluated the comprehensive impact of Cape Town's surface runoff and urbanization on the water system before and after the drought crisis through natural water loss and groundwater recharge.

2.7. Water-related electricity use and GHG emission analysis

Similar to the boundary selection for the water system performance analysis, the system boundary for the energy and GHG emissions analysis is set at the administrative boundary of Cape Town. Data are collected from water infrastructure operated by the City of Cape Town and located in Cape Town but managed by other authorities, such as water supply dams, wastewater treatment facilities, and hydroelectric power stations. Detailed data are shown in Table S2. Because of uncertainty, complexity, and data lags in Cape Town energy and emissions forecasts, this study focused on comparing the situation before and during the drought crisis to determine the environmental impact of urban water services, the possibility of improving water supply and wastewater treatment energy utilization, and the impact of drought on related data. In this case, electricity total (E_t), electricity water-related (E_w), and electricity generated (E_g), respectively, referred to Cape Town's total annual electricity consumption, electricity consumption by water supply and wastewater treatment facilities, and electricity produced by water facilities during the study period. Furthermore, in this study, GHG total (G_t) and GHG water-related (G_w) represented the average annual total energy-related emissions and energy emissions related to water supply and water treatment in Cape Town during the study period, respectively.

In Table 3, water-related electricity utilization indicates annual per capita water-related electricity consumption. According to the formula, the lower the per capita water-related electricity consumption, the higher the efficiency of the water supply system. Water facility electricity consumption, as a percentage of the city's total electricity consumption, indicates the energy dependence of water services, such as water supply, drainage, and wastewater treatment. The contribution of electricity generated by wastewater facilities reflects the ability of water

Table 3
Water-energy nexus indicators.

Indicators	Description	Formulas
Water-related electricity utilization (MJ/capita/annual)	Total water-related electricity use per capita per year	Ew/Pop
Water-related electricity proportion (%)	Water-related electricity/total electricity	$Ew/Et*100\%$
Self-sufficiency of electricity generated by wastewater facilities (%)	Electricity generated by wastewater facilities/water-related electricity	$Eg/Ew*100\%$
Water-related energy GHG (tonnes CO ₂ e/capita/annual)	Total water-related GHG per capita per year	Gw/Pop
Water-related energy GHG proportion (%)	Water-related GHG/total GHG from energy	$Gw/Gt*100\%$

facilities to be self-sufficient in electricity consumption. Moreover, annual water-related energy emissions per capita and water-related energy emissions as a percentage of total greenhouse gases were used to assess the carbon footprint and environmental performance of the urban water system in Cape Town before and during the drought crisis.

3. Data analysis and results

This study analyzed indicators that contributed to the performance

Table 4
Urban water metabolism performance of Cape Town.

Indicators	2009–2012	2015–2018	2020–2023	2037–2040
Rainwater Utilization for Water Supply				
Centralized supply use rate (%)	17.35	22.34	17.62	20.39
Decentralized supply use rate (%)	7.10	6.30	4.33	3.92
Total supply use rate (%)	24.45	28.64	21.95	24.30
Urban Water Metabolism Performance				
Urban water utilization (kL/capita/annual)	95.83	72.26	92.35	80.00
Water supply internalization (%)	28.76	26.03	27.14	22.49
Wastewater potential for water supply	0.79	0.65	0.62	0.58
Natural Loss Potential for water supply	2.54	1.97	2.37	1.67
Hydrological Performance				
Hydrological performance compared with drought period-Natural loss	1.46	NA	1.71	1.32
Hydrological performance compared with drought period-Groundwater recharge	2.16	NA	2.44	1.97
Water-related Energy Use				
Water-related electricity utilization (MJ/capita/annual)	186.53	170.36	NA	NA
Water-related electricity proportion (%)	1.46	1.67	NA	NA
Self-sufficiency of electricity generated by wastewater facilities (%)	12.2	18.87	NA	NA
Water-related energy GHG (tonnes CO ₂ e/capita/annual)	0.09	0.07	NA	NA
Water-related energy GHG proportion (%)	1.56	1.56	NA	NA

of urban water metabolism over four distinct periods. The results of these indicators are shown in Table 4. The results of the data analysis included both the trends observed in performance metrics and the examination of underlying causes that may account for these observed trends.

3.1. Water metabolism characteristics from 2009 to 2012

In Fig. 4, from 2009 to 2012, the material flow analysis (MFA) indicates that the centralized water supply system was the predominant source of industrial and residential water used in Cape Town. In the absence of uncontrollable and unstable factors such as climate change and the pandemic, the internalization of the water system reached 28.76%. Regarding the potential for wastewater as part of the water supply, the indicator was at its peak (0.79) throughout the study periods, which denoted a significant volume of wastewater that could have been repurposed for use but was not utilized in practice. Saldías et al. (2016) estimated that wastewater reuse constituted less than 5% of South Africa's national water systems in 2010, suggesting a consistently low reuse rate across the country, including Cape Town. This could likely be attributed to policy gaps and technological limitations at that time, as well as low public acceptance and trust, as evidenced by some resistance and skepticism regarding the safety and suitability of using recycled wastewater, especially for drinking purposes (Saldías et al., 2016).

The natural loss potential for the water supply also peaked during four research periods, which more than doubled (2.54 times) that of centralized and decentralized water supply systems. This result indicated that a substantial amount of surface runoff was not utilized. A contributing factor to this underutilization may have been the shortage of certain ground infrastructures, such as rain gardens, which emerged as a water conservation strategy in South Africa around the mid-2010s, a period marked by escalating water scarcity challenges (City of Cape Town, 2019). A portion of the surface runoff is retained in these surface infrastructures and is subsequently filtered and replenished into the urban water supply network, surface water bodies, and underground water bodies. The above results and phenomena highlight the under exploitation of alternative water sources in Cape Town between 2009 and 2012, as well as the need to develop integrated urban water management strategies to overcome potential water supply uncertainties. Furthermore, regarding the utilization of rainwater in the water supply, the peak decentralized supply use rate was recorded at 7.10%. This peak was primarily attributable to the insufficient coverage and imperfections in the centralized water supply system at that time, which required a considerable reliance on community-based supply systems (Turpie and Letley, 2023).

3.2. Water metabolism characteristics from 2015 to 2018

In 2015, Cape Town began to experience its worst period of drought since 1904, and extreme water scarcity began to threaten the supply of freshwater for its citizens (OECD, 2021). As shown in the MFA in Fig. 5, there was a considerable decline in annual precipitation by roughly 46.4% from the baseline period of 2009–2012, which was concurrent with a population surge of 16.7%. Urban water utilization was also greatly impacted, dropping from 95.83 kL/capita/annual to 72.26 kL/capita/annual. To address this hydrological deficit, Cape Town's governance implemented diverse water conservation strategies, including public education campaigns, network pressure management, and water-efficient technologies (Calverley and Walther, 2022).

In addition, more non-traditional water sources were utilized through increased infrastructure investment, such as groundwater abstraction, water recycling, and rainwater collection. This led to a reduction in the total rate of water loss (combined with wastewater and natural losses) from 3.33 to 2.62, representing a substantial decrease of approximately 21%. As a unique water source for a coastal city, Cape Town is also actively exploring the possibility of desalination as an

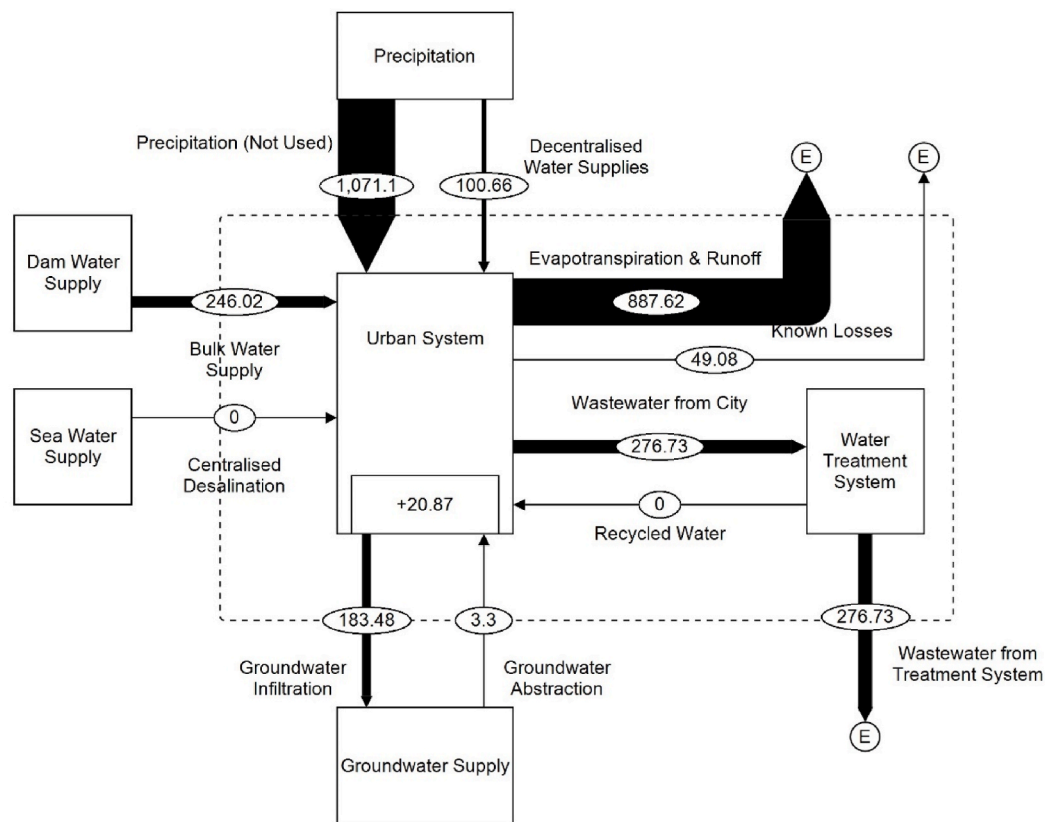
Import: $\sum$ Import Tg/adStock: Δ Stock Tg/aExport: $\sum$ Export Tg/a

Fig. 4. Water metabolism of Cape Town from 2009 to 2012 (unit: GL/annual).

alternative water source during this period (City of Cape Town, 2019). In the medium and longer term, desalination can be an increasingly significant water source because it is scalable and not dependent on rainfall (Prado de Nicolás et al., 2023). Another important indicator of progress was the increase in rainfall utilization from 24.45% to 28.64% from 2009 to 2012. In addition, water supply internalization decreased by 2.73% following the implementation of water allocation guidelines during drought conditions (City of Cape Town, 2019).

The period from 2015 to 2018 was marked by low rainfall and increasing water demands from the city's rapidly growing population, which strained water resources and, consequently, impacted water-related energy use due to the need for more intensive water supply and treatment measures. The good news is that water consumption in the energy sector has been effectively controlled thanks to the implementation and guidance of the Integrated Resource Plan (IRP) 2010–2030 (Department of Mineral Resources and Energy, 2024). However, environmental and demographic shifts have inevitably impacted water-related electricity utilization, as evidenced by a decrease from 186.53 to 170.36 MJ/capita/annual, a reduction of 10% from the preceding period. At the same time, GHG emissions associated with water facilities fell by 0.02 tons of CO_{2e} per person per year. Moreover, the use of solar panels in wastewater treatment facilities has almost doubled the contribution of wastewater treatment facilities to the overall power grid (City of Cape Town, 2021).

3.3. Water metabolism characteristics from 2020 to 2023

During the COVID-19 pandemic, stay-at-home orders and lockdowns increased residential water use as people engaged more frequently in activities such as cooking, cleaning, bathing, and handwashing, which

caused a 10–15% rise in the demand for domestic water (Bera et al., 2022). Furthermore, population growth fueled demand in the manufacturing sector, which maintained the levels of industrial water consumption. The MFA in Fig. 6 shows that the bulk water supply peaked at 281.23 GL/annual, marking the highest level compared with the previous periods. A particularly striking result was that the total rainfall supply use rate plummeted to 21.95%, the lowest of all four study periods. This decrease was attributed to the effects of diminished industrial activities on precipitation collection and management, further exacerbating the pandemic's profound impact on water resource dynamics. In addition, compared with the drought period, the two indicators of hydrological performance reached the highest values among the three periods, showing that Cape Town's natural conditions rapidly recovered from the drought crisis. The changes also reflected the regulating effect of the existence of the ocean on the natural water cycle.

3.4. Water metabolism characteristics from 2037 to 2040

Based on the water system development strategy and water-sensitive visions and goals, the changes in water flows from 2037 to 2040 are shown in the MFA provided in Fig. 7. In this future period, water system internalization is expected to further decrease to 22.49%, suggesting the need for an even more integrated and cohesive approach to water distribution and management. In this scenario, the amount of seawater desalination (54.75 GL/annual) has reached more than 20% of the bulk water supply (260.35 GL/annual). The changes will mainly be in response to the pressures of urbanization and the need to manage water resources more effectively in the face of population growth and climate change stressors.

However, a decrease in the internalization of the water supply did

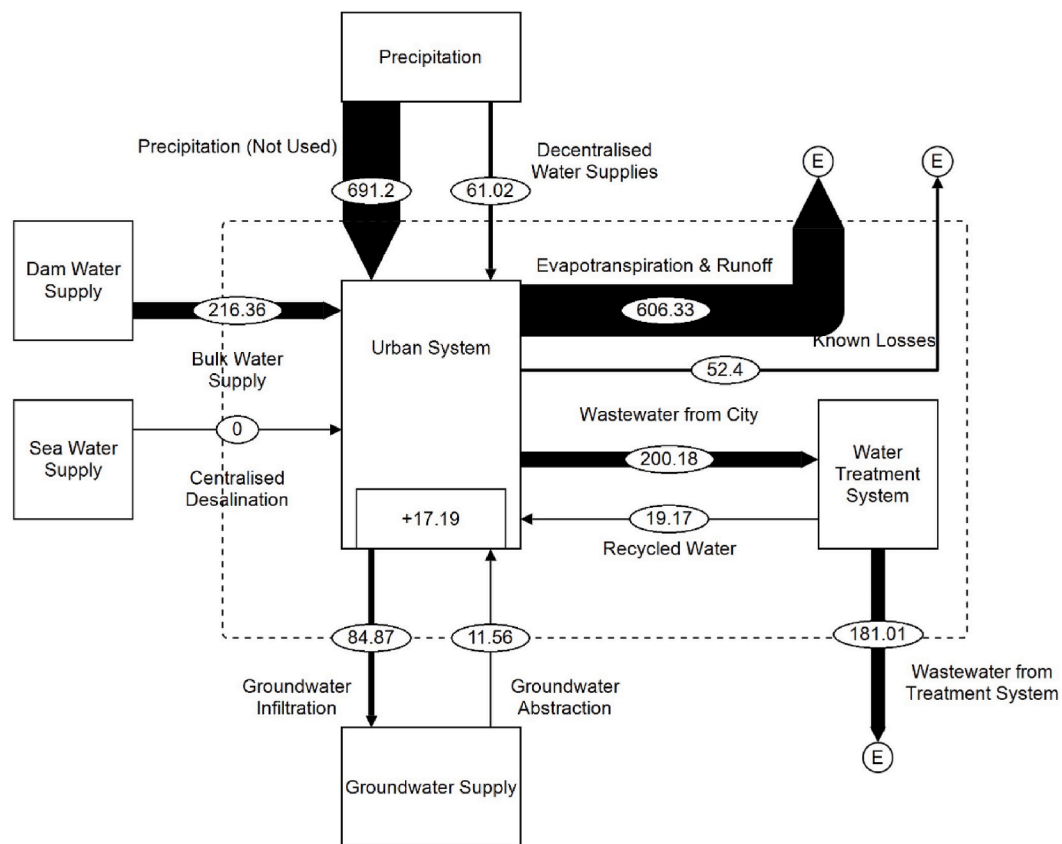
Import: $\sum$ Import Tg/adStock: Δ Stock Tg/aExport: $\sum$ Export Tg/a

Fig. 5. Water metabolism of Cape Town from 2015 to 2018 (unit: GL/annual).

not mean a decrease in the variety of water sources. In the future period from 2037 to 2040, the combined potential for total water supply from wastewater and natural loss, another crucial metric, showed a decrease of 2.25 compared to earlier years. This decline could reflect the maturation of water reuse strategies, which were becoming increasingly optimized and efficient over time, possibly due to the emergence of new technological solutions, increased infrastructure investment, and heightened public awareness of water conservation needs. Moreover, in the context of the overall decrease in precipitation and the standardization of the decentralized water supply in 2040, the overall rainwater utilization rate remained at the level before the drought crisis, which indicated the more efficient use of groundwater and surface runoff in the Cape Town area (City of Cape Town, 2023).

4. Discussion

4.1. Urban water metabolism performance

As shown in Fig. 8, Cape Town's urban water utilization peaked between 2009 and 2012 at 95.83 kL per person per year; however, this result was not positive. According to statistics, per capita water consumption during a drought crisis can be reduced by about one-third of pre-drought levels without affecting the overall national economy (City of Cape Town, 2019). In other words, this phenomenon reflects the huge potential of Cape Town to save water and the application prospects of water-saving facilities. According to Cape Town (2023) Outlook, it has been estimated that by 2040, per capita water savings could exceed 15 kL per person per year, which would be a huge achievement. The considerable decline of more than 23 kL per person per year from 2015

to 2018 compared with the previous year reflects the impact of the drought crisis on Cape Town's water system. In Cape Town's water supply internalization, the decentralized water supply between 2009 and 2012 was more than double the expected 2040 level. This was mainly because of the large amount of non-standardized surface runoff abstracted from temporary and informal settlements.

Before the drought, some communities had only one tap for every 25 households in the formal water supply network (City of Cape Town, 2019). The use of water without treatment not only brings huge risks to the health and welfare of residents but also is inconsistent with Cape Town's safe access to water and sanitation policy (OECD, 2021). The COVID-19 pandemic has also raised the awareness of the people of Cape Town of safety and health issues. Since 2015, Cape Town has made significant investments in existing water infrastructure and the construction of future facilities, such as desalination plants, reaching approximately 500 million ZAR in 2021 (City of Cape Town, 2023). Moreover, under the IRP's guidance to reduce water use in the energy sector, limiting water use in energy facilities has become a national long-term investment strategy (Department of Mineral Resources and Energy, 2024). The investment trend is shown in Fig. 9.

The analysis of Cape Town's hydrological performance before and after the drought crisis revealed that the impact of reduced rainfall on groundwater replenishment was much more significant than that of urban surface runoff and evaporation. It is worth noting that rainfall losses increased following the drought crisis because of the impact of Cape Town's urbanization and hardening of road surfaces. Following the publication of Cape Town's Water Strategy and Water Outlook reports, increasing attention will be paid to the utilization of surface runoff and groundwater, as well as to the water-saving infrastructure such as

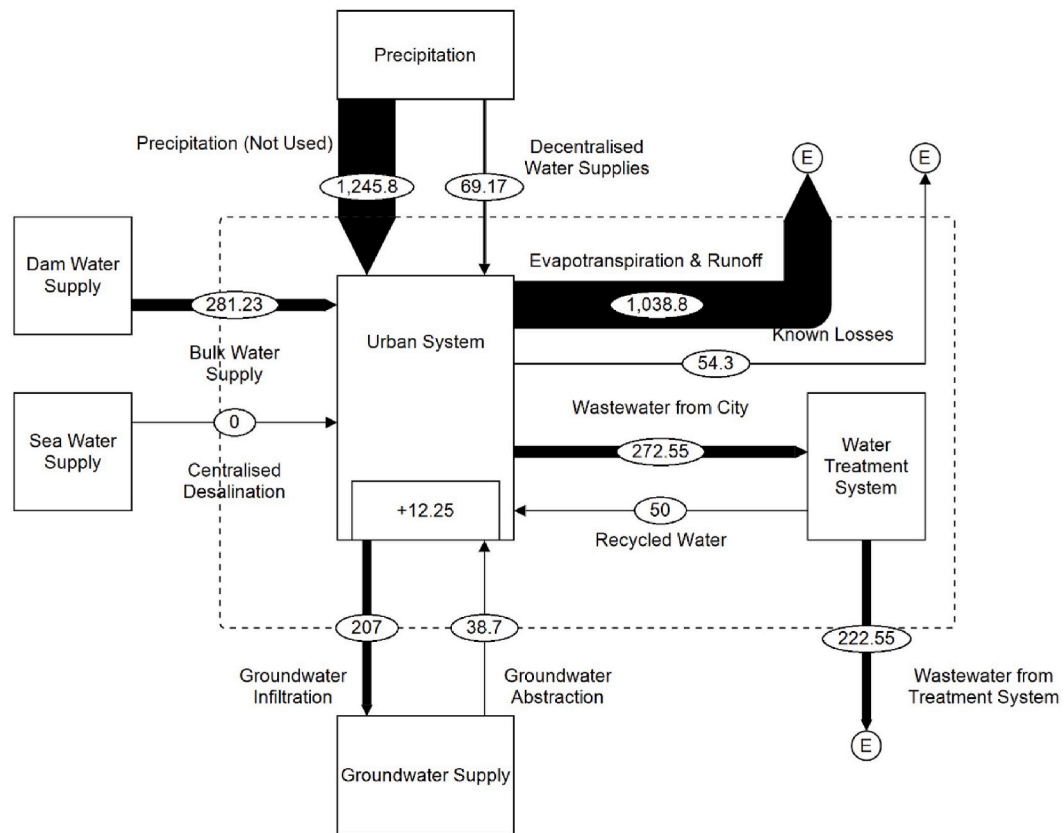
Import: $\sum$ Import Tg/adStock: Δ Stock Tg/aExport: $\sum$ Export Tg/a

Fig. 6. Water metabolism of Cape Town from 2020 to 2023 (unit: GL/annual).

raingardens. Furthermore, as a water-sensitive city, Cape Town aims to increase the use of natural water sources by 2040 and compensate for unavoidable natural losses by activating more coastal desalination facilities. Fig. 10 shows the changes in hydrological performance during the four research periods.

4.2. Measures towards a water-sensitive city

As discussed in Section 3, based on the performance of Cape Town's water system in becoming a water-sensitive city, the city's authorities have implemented or will implement five measures aimed at increasing alternative water resources, reducing water losses, and improving the health of the urban water cycle.

- (1) Cape Town is committed to improving its bulk water supply system. To achieve this goal, Cape Town launched an alien vegetation clearing initiative in 2019, which will increase the supply capacity of existing river channels. Moreover, Cape Town authorities continue to implement plans to extend the Berg River to the Voelvlei Dam. In the future, new surface water schemes are expected to increase the bulk water supply by 100 ML per day (City of Cape Town, 2019). To fulfill their commitment to Wise Use, municipal departments have promoted the use of centralized water supply pipe networks for households (OECD, 2021). Both of these measures can help alleviate water shortages by improving natural water use efficiency and introducing new alternative sources of water supply.
- (2) Cape Town's water authority continues to increase the use of alternative water sources to supplement the centralized water supply system. The currently selected sources are groundwater

extraction and seawater desalination. Groundwater withdrawals from the three aquifers around Cape Town have gradually increased since 2018, reaching 38.7 GL per year in 2023. A desalination facility is expected to be commissioned in 2026, which will provide a water production capacity of 50 ML per day (City of Cape Town, 2023).

- (3) An important part of Cape Town's commitment to Sufficient, Reliable Water from Diverse Sources, a standardized, decentralized water supply system cannot be ignored (City of Cape Town, 2019). Many remote and rural communities urgently need water provided by decentralized facilities, which is also the most economical and convenient option. In this context, the Cape Town government has developed regulatory standards for the operation of decentralized water facilities and the quality of their water output. In addition, residents are restricted from drawing water directly from rivers, lakes, and wells. These measures reduced the rate of decentralized rainwater use from 7.10% to 3.92%. Nevertheless, other alternative water sources, such as desalination, groundwater withdrawal, and wastewater reuse, can compensate for the decline in decentralized rainwater use and improve the health of the urban water metabolism and the well-being of residents.
- (4) Cape Town authorities continue to work with private companies and organizations to reduce surface runoff and evapotranspiration by establishing urban rain gardens, green roofs, and urban waterways (OECD, 2021). Based on the projections for 2037–2040 produced by this study, when Cape Town is expected to become a water-sensitive city, the natural loss potential of water supply is projected to stabilize at less than 2.

Import: $\sum$ Import Tg/a

dStock: Δ Stock Tg/a

Export: $\sum$ Export Tg/a

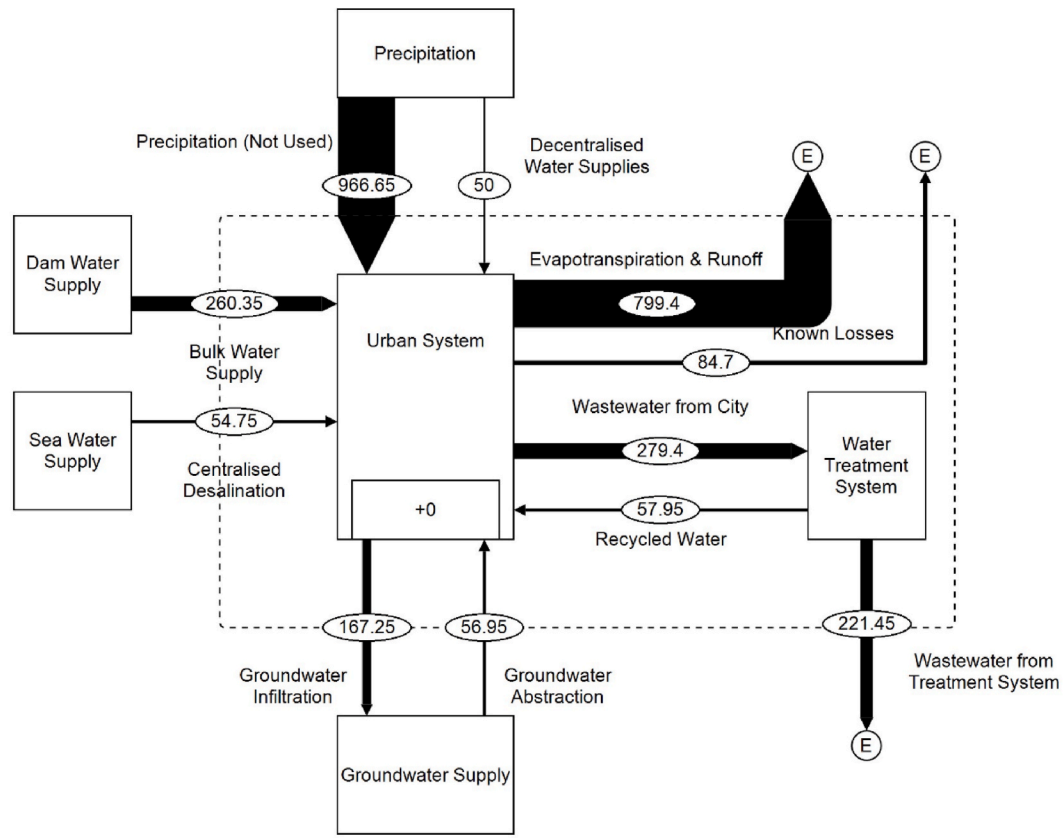


Fig. 7. Water metabolism of Cape Town from 2037 to 2040 (unit: GL/annual).

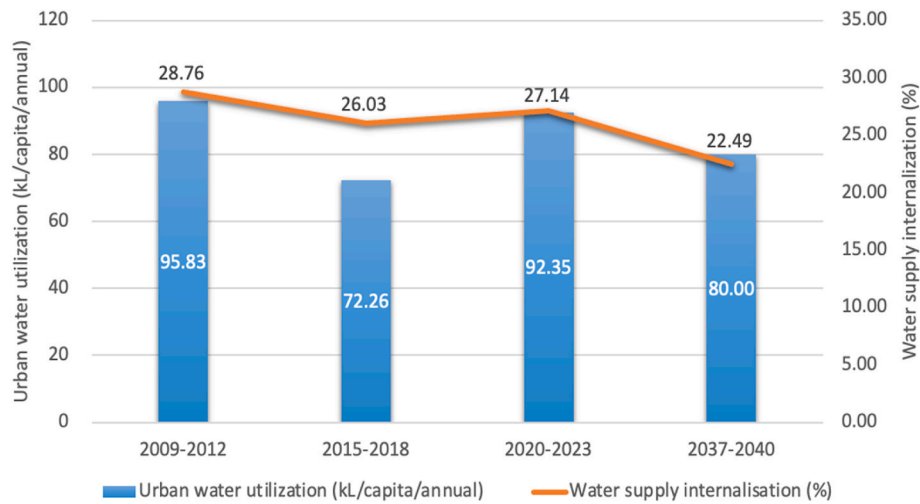


Fig. 8. Trends about urban water utilization and internalization.

(5) The Cape Town authorities and the Nature Conservancy jointly launched the Greater Cape Town Water Fund and a series of other investment plans related to water systems. Recently, related investments have been directed at water recycling in wastewater treatment facilities (City of Cape Town, 2019). Because of these investments, the potential use of wastewater in the water supply has gradually declined since 2009, and it is expected to be 0.58 in

2040. In other words, an increasing amount of water will not enter the wastewater stream for final disposal.

4.3. Water-energy nexus before and during the drought crisis

As shown in Fig. 11, water-related electricity utilization from 2015 to 2018 decreased compared with the utilization from 2009 to 2012 because of a reduction in total water flows from 186.53 MJ to 170.36 MJ

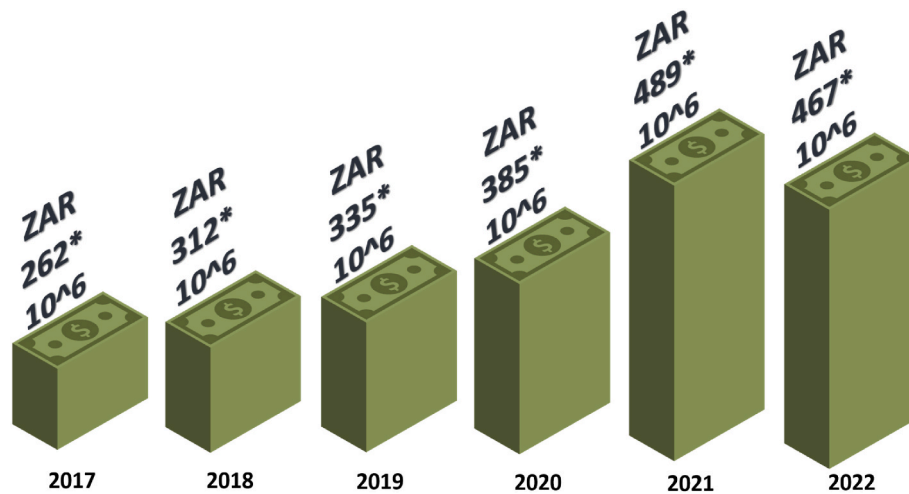


Fig. 9. Cape Town investment trend in water infrastructures and technologies.

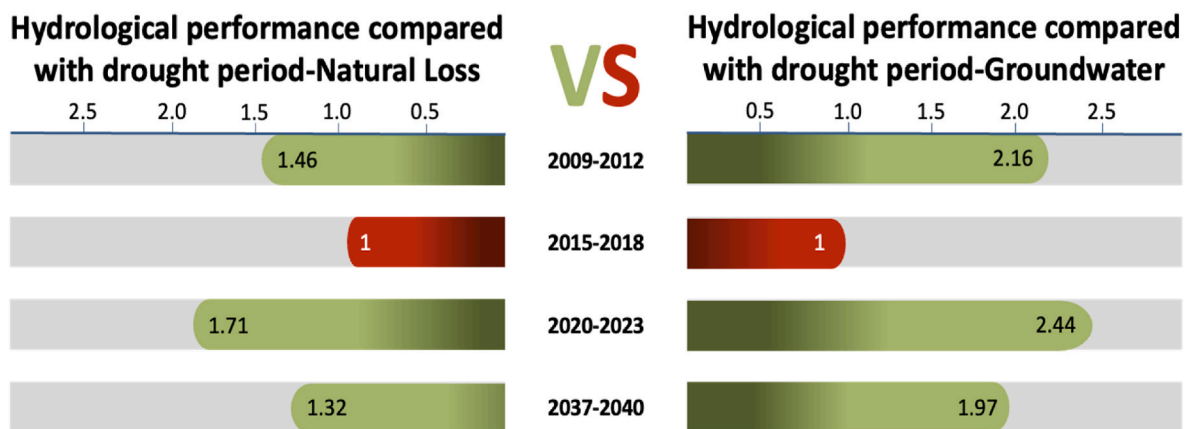


Fig. 10. Trends in two indicators about hydrological performance.

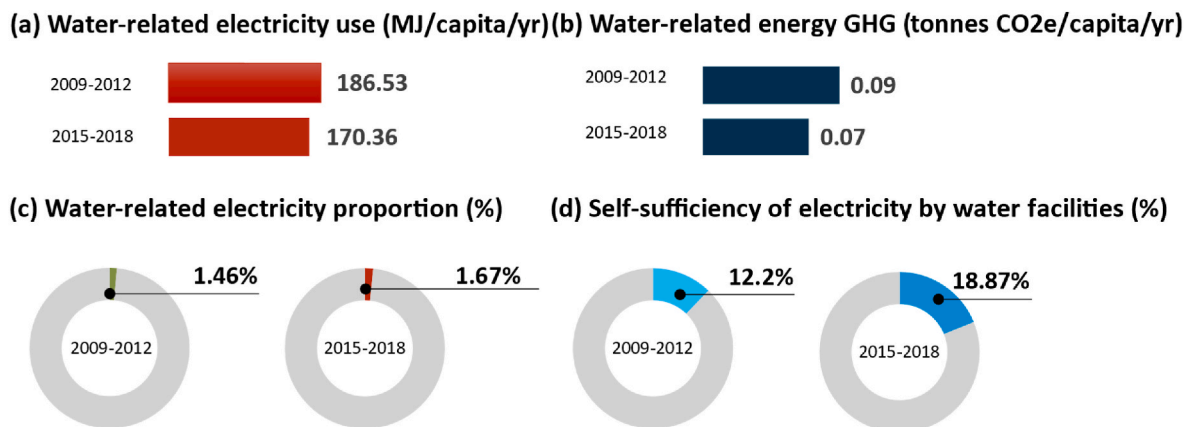


Fig. 11. Indicators about water-energy nexus.

per person per year. However, in comparison, the proportion of water-related electricity consumption in total electricity consumption increased by 0.2%. This increase was due to the severe impact of the drought crisis on Cape Town's industry, resulting in a significant drop in industrial electricity consumption (City of Cape Town, 2021). Because of a significant decrease in the amount of water entering the water facility due to poor terrain, energy production dropped from more than 80,000

GJ per year to 44,000 GJ per year in the period from 2009 to 2012. The installation of solar panels for water facilities increased the contribution of electricity generated by wastewater facilities by 0.14%.

Moreover, according to measurements taken by the electricity department, electricity consumption in some sectors at the end of 2018 increased toward the end of the drought crisis, including domestic hot water use (City of Cape Town, 2021). The scale of the intensive water

treatment processes adopted during this period required varying levels of energy inputs. This result aligns with [Oloade's \(2018\)](#) study on the energy–water interplay in South Africa's coal-fired power plants. On the other hand, it is undeniable that overall water availability in 2015–2018 and the implementation of regulations and investments under the IRP have led to a reduction in water use in the energy sector ([City of Cape Town, 2018](#); [Department of Mineral Resources and Energy, 2024](#)). Regarding GHG emissions, there was a significant decrease (22.2%) in water-related emissions due to a reduction in total water flows. However, as a share of total emissions, water-related emissions did not change because of industrial shutdowns and more complex water acquisition and treatment processes during drought from 2015 to 2018. In the future, vigorously developing hydropower, using more recycled water in the energy sector, and making water consumption a key criterion for long-term investment in the power sector are the main strategies for Cape Town to improve its water-energy nexus.

4.4. Limitations and future works

The estimation of water flows is the first limitation of this study. Single-day or single-month changes in water flows were integrated into annual totals in the four study periods. Hence, some characteristics of natural water flows, such as seasonal variations and concentrated precipitation events, were not included in the data analysis, which may have interfered with the development of water management strategies. The lack of a precise analysis of wet and dry seasons could affect the dynamic adjustment of the water supply system ([Cai et al., 2020](#)). In this study, the water balance model was simplified based on Cape Town's situation and the available data. Hence, some simplified flows may have obscured a specific situation ([Vazifeh et al., 2021](#)). For example, because of the lack of available data, evapotranspiration and runoff were merged into a single flow. Regarding anthropogenic flows, water flows that interacted with various other material flows, such as nutrients and waste, were not considered, although these interactions can be complex and crucial. Another limitation of this study is the limited availability of data on the water–energy nexus. Because Cape Town is a continuously growing city, its energy system projections are hindered by challenges and uncertainties.

For future research about the metabolic performance, several influential factors can be further considered, such as housing density, land-use distribution, and ocean impacts, which increased the accuracy and credibility of the assessment results. Previous studies have demonstrated the suboptimal water supplies faced by families at various income levels in Cape Town, recommending improvements that can be considered in future research ([Cook et al., 2021](#); [Strydom et al., 2020](#); [Wang et al., 2024](#)). In addition, future research could focus on the temporal and spatial effects of climate change on water flows in Cape Town. Such studies could provide valuable guidance for Cape Town's transition to a water-sensitive city and make the analysis results more comprehensive and accurate.

5. Conclusions

The results of this study provide insights into the intricate interactions between natural and anthropogenic water flows in coastal areas based on the water balance model and water metabolism assessment framework. Based on the results, this study offers a nuanced understanding of Cape Town's water resource management. It examined the dynamics of coastal urban water flow under varying conditions influenced by climate change, drought, and the COVID-19 pandemic. Furthermore, it emphasized the importance of integrating energy considerations into the water management narrative to establish a water–energy nexus that reflects the interconnectedness of coastal urban systems. These results indicate that Cape Town needs to adopt a water-sensitive approach that aligns with global sustainability goals and local socioeconomic realities. In addition, a strategic blend of technological

innovation, infrastructural upgrades, and policy reforms is needed to optimize water usage, reduce dependency on conventional water sources, and mitigate environmental impacts. The results indicate the pressing need to diversify water sources, enhance water recycling and reuse, and adopt green and desalination infrastructure to bolster the sustainability of Cape Town's urban water cycle. In summary, this comprehensive analysis provides valuable insights into Cape Town's urban water metabolism and sets a precedent for other coastal cities that grapple with similar water-related challenges. Based on the results of this study, an integrated, forward-looking coastal urban water management approach can be further developed to emphasize the symbiosis between water sustainability, energy efficiency, and urban resilience.

CRediT authorship contribution statement

Ziyu Wang: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Zehua Lu:** Writing – original draft, Investigation, Formal analysis. **Jiaqi Ma:** Writing – original draft, Investigation. **Guohe Huang:** Writing – review & editing. **Chunjiang An:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ocecoaman.2024.107434>.

Data availability

The data supporting this article have been included in the article and the Supplementary Material.

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